

Microsoft Foundry Embedding Model Recommendation

Recommendation for a Corporate Policy RAG knowledge base using the embedding models available in the project region

Executive Recommendation

Start with Cohere-embed-v3-english. For an English-language Corporate Policy Assistant, it provides a strong balance of retrieval quality, operational simplicity, and fit for semantic search and Retrieval-Augmented Generation (RAG).

If you expect multilingual documents, image-rich documents, or multimodal retrieval in the near future, **embed-v-4-0** becomes the stronger strategic choice. If you want an open-model alternative for experimentation and portability, **qwen--qwen3-embedding-0.6b** is a good development candidate.

1. Best Starting Choice: Cohere-embed-v3-english

This model is the best practical fit when your policy documents and user questions are primarily in English. It is well suited to semantic search, document retrieval, question-to-passage matching, and RAG workflows.

- Optimized for English-language semantic retrieval.
- Appropriate for enterprise knowledge-base and RAG scenarios.
- A mature embedding option that avoids deploying an unnecessarily large open model.
- Simpler operational choice than very large 8-billion-parameter embedding models.
- Well suited to matching a user question with a policy passage even when the wording is different.

Example semantic match

User question: “How many vacation days can an employee carry over?”

Relevant policy text: “Employees may transfer a maximum of 40 unused paid-time-off hours into the following calendar year.”

The embedding model should recognize that “vacation days,” “carry over,” “unused paid-time-off,” and “transfer into the following calendar year” express closely related meaning even though the words are not identical.

Recommended RAG Flow

```
Corporate policy documents
↓
Chunk documents
↓
Cohere-embed-v3-english
↓
Azure AI Search vector index
↓
User question
↓
Retrieve matching policy passages
↓
LLM generates a grounded answer
```

2. Second Choice: embed-v-4-0

Choose **embed-v-4-0** when your requirements extend beyond straightforward English text retrieval. It is especially attractive if the solution may need broader multilingual or multimodal retrieval capabilities.

Consider Embed 4 when you expect:

- Policies in multiple languages.
- Scanned pages or images that may participate in retrieval.
- Tables, diagrams, or other visually meaningful policy content.
- A future knowledge base combining text and images.
- A desire to standardize on a newer generation of Cohere embedding technology.

For a normal English text-policy RAG implementation, these additional capabilities may be more than you need initially. A newer model is not automatically the better choice if it adds cost or complexity without improving retrieval on your actual policy corpus.

3. Best Open-Model Alternative: qwen--qwen3-embedding-0.6b

This is a strong candidate when you want a Hugging Face/open-model option with a smaller operational footprint than the 8B variant.

- Lower compute requirements than the 8B Qwen3 embedding model.
- Lower expected latency and easier development deployment.
- Useful multilingual capability.
- Greater portability and less dependence on a single managed model provider.
- A practical benchmark candidate against the Cohere models.

I would not start with qwen--qwen3-embedding-8b. The larger model may provide quality advantages in some workloads, but it is likely excessive for a Phase 1 policy RAG system unless testing proves the additional retrieval quality justifies the compute, latency, and operating cost.

4. Other Viable Models

snowflake--snowflake-arctic-embed-l-v2.0

A strong retrieval-oriented open model and a good benchmark candidate, particularly if multilingual retrieval or open-weight deployment is important.

intfloat--e5-large-v2

A respected English embedding model. It is useful as a benchmark, but it typically benefits from correct query and passage prefixes, so implementation details matter.

sentence-transformers--all-minilm-l6-v2

Very useful for low-cost prototypes, small datasets, and fast CPU inference. It is better viewed as a speed-and-cost option than a preferred enterprise production model.

sentence-transformers--msmarco-bert-base-dot-v5

Designed for passage retrieval and still viable, but newer retrieval models are generally better candidates for a new implementation.

NVIDIA Nemotron 1B / 8B embedding models

Potentially strong models, especially in NVIDIA-oriented environments. They become more attractive if Foundry pricing, capacity, or measured retrieval quality makes them competitive; otherwise they may add unnecessary complexity for an initial policy RAG implementation.

5. Recommended Ranking for Your Project

Rank	Model	Recommended Use
1	Cohere-embed-v3-english	Best starting point for English corporate-policy RAG
2	embed-v-4-0	Best when multilingual or multimodal retrieval is expected
3	Qwen3-Embedding-0.6B	Best open-model development alternative
4	Snowflake Arctic Embed L v2.0	Strong open multilingual retrieval candidate
5	E5-large-v2	Good English baseline; implementation prefixes matter
6	all-MiniLM-L6-v2	Low-cost prototype rather than preferred production model

6. Suggested Evaluation Approach

After establishing a working baseline with Cohere-embed-v3-english, compare it with qwen--qwen3-embedding-0.6b using approximately 25-50 representative policy questions.

Measure:

- Whether the correct policy appears in the top three retrieved results.
- Whether the most relevant paragraph appears as the first result.
- Retrieval latency.
- Endpoint and indexing cost.
- Quality with corporate acronyms and internal terminology.
- Performance when the user question is phrased differently from the source policy text.

The best embedding model is not necessarily the model with the highest public benchmark score. The best model is the one that retrieves **your actual corporate-policy passages** most reliably and economically.

7. Important Azure AI Search Design Consideration

Once policy chunks have been embedded and stored in Azure AI Search, changing embedding models generally means generating new vectors and rebuilding or updating the vector index. The vector field dimensions configured in Azure AI Search must match the dimensions produced by the selected embedding model.

For that reason, it is worth completing a small retrieval evaluation before indexing the full policy corpus.

Final Recommendation

Phase 1: Use **Cohere-embed-v3-english** for the initial Corporate Policy RAG implementation.

Benchmark: Compare it against **qwen--qwen3-embedding-0.6b** using representative policy questions.

Future expansion: Consider **embed-v-4-0** if multilingual, image, or multimodal retrieval becomes a requirement.